

Report: Analysis of NHANES data

Objectives

- Can the presence of interactions- and other non-linear effects be inferred by comparing the prediction performance of purely linear base model with random forest base models in MVS?
- Can residualizing the outcome with respect to a linear model help to evaluate non-linear or interaction signal using random forests?

Summary & Conclusion

- In the empirical example using the NHANES data, random forest base models in the MVS configuration showed **no improvement** in predictive performance compared with MVS using only linear models. Therefore, coefficients are not further interpreted.

Data

For the demonstration of the MVS extension, we used multi-source data from the US National Health and Nutrition Examination Survey (NHANES). The covariates are organized into six groups and a sum score of a depression questionnaire is used as the outcome. The sample consists of young adults aged between 20 and 30 years.

The data were selected based on age including participants aged between 20 to 30 years with a sample size of 853. Variables were selected based on available data. Descriptions of the variables are listed in a later section.

The group labels are:

- sociodemographics,
- dietary behavior,
- medical conditions,

- physical activity,
- sleep, and
- body measurements.

Analysis

The analysis compares three different MVS configurations. Two configurations use consistent base learner, either linear base models or random forest base models only. The third configuration uses residual models, with linear base models fitted to the groups of covariates in views 1 to 6 and random forest models fitted to the same groups of covariates in views 7 to 12 using a residualized outcome obtained by subtracting the predictions from a linear model fitted to the corresponding group of covariates. 10-fold cross-validation was used for evaluating predictive performance.

Results

Table 1

Predictive performance

	MVS Linear	MVS Residual Models
MSE (SD)	11.8 (0.01)	11.8 (0.02)
Test R^2	0.22	0.22

Note. The variance of the outcome in the complete data is 15.2. Predictive R^2 was calculated as $1 - \text{mse} / \text{var}(y_{\text{test}})$. The reported values are calculated as means.

Table 2

Medians of standardized coefficients

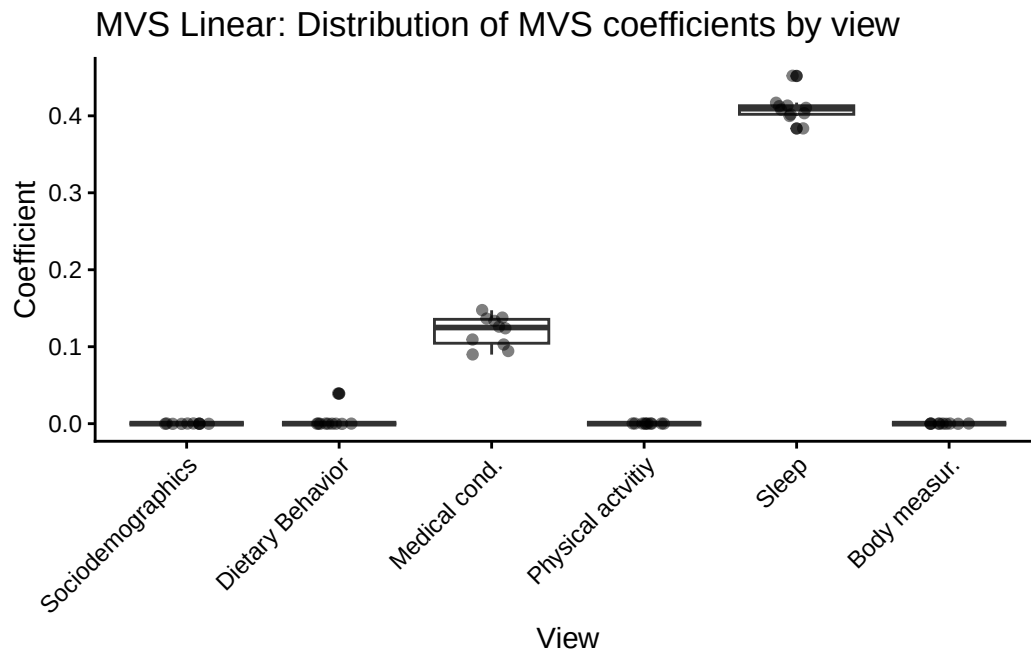
	MVS Residual Models		
	MVS Linear	Linear	Random Forest
Sociodemographics	0	0	0
Dietary behavior	0	0	0.01
Medical condition	0.13	0.11	0
Physical activity	0	0	0
Sleep	0.41	0.41	0
Body measurements	0	0	0

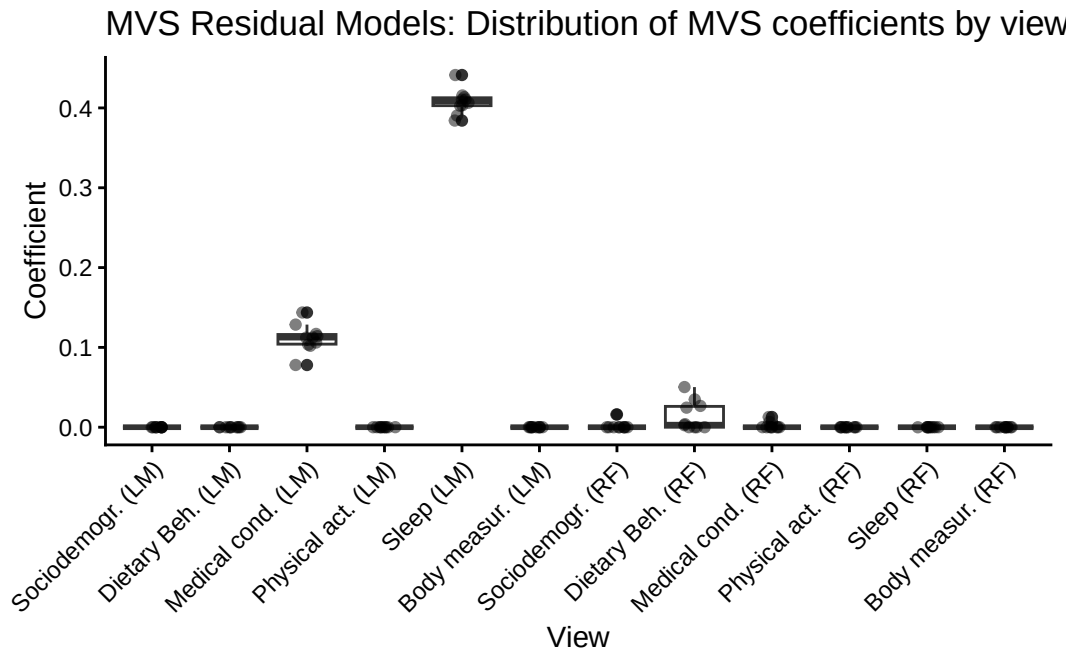
Plots of coefficient distributions

Attaching package: 'ggplot2'

The following object is masked from 'package:randomForest':

margin





Variable names and descriptions

	Label	Variable name	Variable description
1	Sociodemographics	RIAGENDR	Gender
2	Sociodemographics	RIDAGEYR	Age in years at screening
3	Sociodemographics	RIDRETH1	Race/Hispanic origin
4	Sociodemographics	RIDRETH3	Race/Hispanic origin w/ NH Asian
5	Sociodemographics	RIDEXMON	Six month time period
6	Sociodemographics	DMDBORN4	Country of birth
7	Sociodemographics	DMDEDUC2	Education level - Adults 20+
8	Sociodemographics	DMDMARTZ	Marital status
9	Sociodemographics	SIALANG	Language of SP Interview
10	Sociodemographics	FIALANG	Language of Family Interview
11	Sociodemographics	FIAPROXY	Proxy used in Family Interview?
12	Sociodemographics	FIAINTRP	Interpreter used in Family Interview?
13	Sociodemographics	MIALANG	Language of MEC Interview
14	Sociodemographics	AIALANGA	Language of ACASI Interview
15	Sociodemographics	WTINTPRP	Full sample interview weight
16	Sociodemographics	WTMECPRP	Full sample MEC exam weight
17	Sociodemographics	SDMVPSU	Masked variance pseudo-PSU

18	Sociodemographics	SDMVSTRA	Masked variance pseudo-stratum
19	Sociodemographics	INDFMPIR	Ratio of family income to poverty
20	Sociodemographics	INDFMMPC	Family monthly poverty level category
21	Dietary behavior	WTDRD1PP	Dietary day one sample weight
22	Dietary behavior	WTDR2DPP	Dietary two-day sample weight
23	Dietary behavior	DR1EXMER	Interviewer ID code
24	Dietary behavior	DRDINT	Number of days of intake
25	Dietary behavior	DR1DBIH	# of days b/w intake and HH interview
26	Dietary behavior	DR1DAY	Intake day of the week
27	Dietary behavior	DBQ095Z	Type of table salt used
28	Dietary behavior	DRQSPREP	Salt used in preparation?
29	Dietary behavior	DR1STY	Salt used at table yesterday?
30	Dietary behavior	DRQSDIET	On special diet?
31	Dietary behavior	DR1TNUMF	Number of foods/beverages reported
32	Dietary behavior	DR1TKCAL	Energy (kcal)
33	Dietary behavior	DR1TPROT	Protein (gm)
34	Dietary behavior	DR1TCARB	Carbohydrate (gm)
35	Dietary behavior	DR1TSUGR	Total sugars (gm)
36	Dietary behavior	DR1TFIBE	Dietary fiber (gm)
37	Dietary behavior	DR1TTFAT	Total fat (gm)
38	Dietary behavior	DR1TSFAT	Total saturated fatty acids (gm)
39	Dietary behavior	DR1TMFAT	Total monounsaturated fatty acids (gm)
40	Dietary behavior	DR1TPFAT	Total polyunsaturated fatty acids (gm)
41	Dietary behavior	DR1TCHOL	Cholesterol (mg)
42	Dietary behavior	DR1TATOC	Vitamin E as alpha-tocopherol (mg)
43	Dietary behavior	DR1TATOA	Added alpha-tocopherol (Vitamin E) (mg)
44	Dietary behavior	DR1TRET	Retinol (mcg)
45	Dietary behavior	DR1TVARA	Vitamin A, RAE (mcg)
46	Dietary behavior	DR1TACAR	Alpha-carotene (mcg)
47	Dietary behavior	DR1TBCAR	Beta-carotene (mcg)
48	Dietary behavior	DR1TCRYP	Beta-cryptoxanthin (mcg)
49	Dietary behavior	DR1TLYCO	Lycopene (mcg)
50	Dietary behavior	DR1TLZ	Lutein + zeaxanthin (mcg)
51	Dietary behavior	DR1TVB1	Thiamin (Vitamin B1) (mg)
52	Dietary behavior	DR1TVB2	Riboflavin (Vitamin B2) (mg)
53	Dietary behavior	DR1TNIAC	Niacin (mg)
54	Dietary behavior	DR1TVB6	Vitamin B6 (mg)
55	Dietary behavior	DR1TFOLA	Total folate (mcg)
56	Dietary behavior	DR1TFA	Folic acid (mcg)
57	Dietary behavior	DR1TFF	Food folate (mcg)
58	Dietary behavior	DR1TFDFE	Folate, DFE (mcg)
59	Dietary behavior	DR1TCHL	Total choline (mg)
60	Dietary behavior	DR1TVB12	Vitamin B12 (mcg)

61	Dietary behavior	DR1TB12A	Added vitamin B12 (mcg)
62	Dietary behavior	DR1TVC	Vitamin C (mg)
63	Dietary behavior	DR1TVD	Vitamin D (D2 + D3) (mcg)
64	Dietary behavior	DR1TVK	Vitamin K (mcg)
65	Dietary behavior	DR1TCALC	Calcium (mg)
66	Dietary behavior	DR1TPHOS	Phosphorus (mg)
67	Dietary behavior	DR1TMAGN	Magnesium (mg)
68	Dietary behavior	DR1TIRON	Iron (mg)
69	Dietary behavior	DR1TZINC	Zinc (mg)
70	Dietary behavior	DR1TCOPP	Copper (mg)
71	Dietary behavior	DR1TSODI	Sodium (mg)
72	Dietary behavior	DR1TPOTA	Potassium (mg)
73	Dietary behavior	DR1TSELE	Selenium (mcg)
74	Dietary behavior	DR1TCAFF	Caffeine (mg)
75	Dietary behavior	DR1TTHEO	Theobromine (mg)
76	Dietary behavior	DR1TALCO	Alcohol (gm)
77	Dietary behavior	DR1TMOIS	Moisture (gm)
78	Dietary behavior	DR1TS040	SFA 4:0 (Butanoic) (gm)
79	Dietary behavior	DR1TS060	SFA 6:0 (Hexanoic) (gm)
80	Dietary behavior	DR1TS080	SFA 8:0 (Octanoic) (gm)
81	Dietary behavior	DR1TS100	SFA 10:0 (Decanoic) (gm)
82	Dietary behavior	DR1TS120	SFA 12:0 (Dodecanoic) (gm)
83	Dietary behavior	DR1TS140	SFA 14:0 (Tetradecanoic) (gm)
84	Dietary behavior	DR1TS160	SFA 16:0 (Hexadecanoic) (gm)
85	Dietary behavior	DR1TS180	SFA 18:0 (Octadecanoic) (gm)
86	Dietary behavior	DR1TM161	MFA 16:1 (Hexadecenoic) (gm)
87	Dietary behavior	DR1TM181	MFA 18:1 (Octadecenoic) (gm)
88	Dietary behavior	DR1TM201	MFA 20:1 (Eicosenoic) (gm)
89	Dietary behavior	DR1TM221	MFA 22:1 (Docosenoic) (gm)
90	Dietary behavior	DR1TP182	PFA 18:2 (Octadecadienoic) (gm)
91	Dietary behavior	DR1TP183	PFA 18:3 (Octadecatrienoic) (gm)
92	Dietary behavior	DR1TP184	PFA 18:4 (Octadecatetraenoic) (gm)
93	Dietary behavior	DR1TP204	PFA 20:4 (Eicosatetraenoic) (gm)
94	Dietary behavior	DR1TP205	PFA 20:5 (Eicosapentaenoic) (gm)
95	Dietary behavior	DR1TP225	PFA 22:5 (Docosapentaenoic) (gm)
96	Dietary behavior	DR1TP226	PFA 22:6 (Docosahexaenoic) (gm)
97	Dietary behavior	DR1_300	Compare food consumed yesterday to usual
98	Dietary behavior	DR1_320Z	Total plain water drank yesterday (gm)
99	Dietary behavior	DR1_330Z	Total tap water drank yesterday (gm)
100	Dietary behavior	DR1BWATZ	Total bottled water drank yesterday (gm)
101	Dietary behavior	DRD340	Shellfish eaten during past 30 days
102	Dietary behavior	DRD360	Fish eaten during past 30 days
103	Medical condition	MCQ010	Ever been told you have asthma

104	Medical condition	MCQ053	Taking treatment for anemia/past 3 mos
105	Medical condition	MCQ080	Doctor ever said you were overweight
106	Medical condition	MCQ092	Ever receive blood transfusion
107	Medical condition	MCQ160A	Doctor ever said you had arthritis
108	Medical condition	MCQ160M	Ever told you had thyroid problem
109	Medical condition	MCQ160P	Ever told you had COPD, emphysema, ChB
110	Medical condition	MCQ160L	Ever told you had any liver condition
111	Medical condition	MCQ520	Abdominal pain during past 12 months?
112	Medical condition	MCQ550	Has DR ever said you have gallstones
113	Medical condition	MCQ560	Ever had gallbladder surgery?
114	Medical condition	MCQ300B	Close relative had asthma?
115	Medical condition	MCQ300C	Close relative had diabetes?
116	Medical condition	MCQ300A	Close relative had heart attack?
117	Medical condition	MCQ366A	Doctor told you to control/lose weight
118	Medical condition	MCQ366B	Doctor told you to exercise
119	Medical condition	MCQ366C	Doctor told you to reduce salt in diet
120	Medical condition	MCQ366D	Doctor told you to reduce fat/calories
121	Medical condition	MCQ371A	Are you now controlling or losing weight
122	Medical condition	MCQ371B	Are you now increasing exercise
123	Medical condition	MCQ371C	Are you now reducing salt in diet
124	Medical condition	MCQ371D	Are you now reducing fat in diet
125	Sleep	SLD012	Sleep hours - weekdays or workdays
126	Sleep	SLD013	Sleep hours - weekends
127	Sleep	SLQ030	How often do you snore?
128	Sleep	SLQ040	How often do you snort or stop breathing
129	Sleep	SLQ050	Ever told doctor had trouble sleeping?
130	Sleep	SLQ120	How often feel overly sleepy during day?
131	Physical activity	PAQ605	Vigorous work activity
132	Physical activity	PAQ620	Moderate work activity
133	Physical activity	PAQ635	Walk or bicycle
134	Physical activity	PAQ650	Vigorous recreational activities
135	Physical activity	PAQ665	Moderate recreational activities
136	Physical activity	PAD680	Minutes sedentary activity
137	Body measures	BMXWT	Weight (kg)
138	Body measures	BMXHT	Standing Height (cm)
139	Body measures	BMXBMI	Body Mass Index (kg/m**2)
140	Body measures	BMXLEG	Upper Leg Length (cm)
141	Body measures	BMXARML	Upper Arm Length (cm)
142	Body measures	BMXARMC	Arm Circumference (cm)
143	Body measures	BMXWAIST	Waist Circumference (cm)
144	Body measures	BMXHIP	Hip Circumference (cm)
	Scale type		
1	categorical		

2	ordinal / continuous
3	categorical
4	categorical
5	categorical
6	categorical
7	ordinal / continuous
8	categorical
9	categorical
10	categorical
11	categorical
12	categorical
13	categorical
14	categorical
15	ordinal / continuous
16	ordinal / continuous
17	categorical
18	ordinal / continuous
19	ordinal / continuous
20	ordinal / continuous
21	ordinal / continuous
22	ordinal / continuous
23	ordinal / continuous
24	categorical
25	ordinal / continuous
26	ordinal / continuous
27	categorical
28	ordinal / continuous
29	categorical
30	categorical
31	ordinal / continuous
32	ordinal / continuous
33	ordinal / continuous
34	ordinal / continuous
35	ordinal / continuous
36	ordinal / continuous
37	ordinal / continuous
38	ordinal / continuous
39	ordinal / continuous
40	ordinal / continuous
41	ordinal / continuous
42	ordinal / continuous
43	ordinal / continuous
44	ordinal / continuous

45 ordinal / continuous
46 ordinal / continuous
47 ordinal / continuous
48 ordinal / continuous
49 ordinal / continuous
50 ordinal / continuous
51 ordinal / continuous
52 ordinal / continuous
53 ordinal / continuous
54 ordinal / continuous
55 ordinal / continuous
56 ordinal / continuous
57 ordinal / continuous
58 ordinal / continuous
59 ordinal / continuous
60 ordinal / continuous
61 ordinal / continuous
62 ordinal / continuous
63 ordinal / continuous
64 ordinal / continuous
65 ordinal / continuous
66 ordinal / continuous
67 ordinal / continuous
68 ordinal / continuous
69 ordinal / continuous
70 ordinal / continuous
71 ordinal / continuous
72 ordinal / continuous
73 ordinal / continuous
74 ordinal / continuous
75 ordinal / continuous
76 ordinal / continuous
77 ordinal / continuous
78 ordinal / continuous
79 ordinal / continuous
80 ordinal / continuous
81 ordinal / continuous
82 ordinal / continuous
83 ordinal / continuous
84 ordinal / continuous
85 ordinal / continuous
86 ordinal / continuous
87 ordinal / continuous

88 ordinal / continuous
89 ordinal / continuous
90 ordinal / continuous
91 ordinal / continuous
92 ordinal / continuous
93 ordinal / continuous
94 ordinal / continuous
95 ordinal / continuous
96 ordinal / continuous
97 ordinal / continuous
98 ordinal / continuous
99 ordinal / continuous
100 ordinal / continuous
101 categorical
102 categorical
103 categorical
104 categorical
105 categorical
106 categorical
107 categorical
108 categorical
109 categorical
110 categorical
111 categorical
112 categorical
113 categorical
114 categorical
115 categorical
116 categorical
117 categorical
118 categorical
119 categorical
120 categorical
121 categorical
122 categorical
123 categorical
124 categorical
125 ordinal / continuous
126 ordinal / continuous
127 ordinal / continuous
128 ordinal / continuous
129 categorical
130 ordinal / continuous

```
131         categorical
132         categorical
133         categorical
134         categorical
135         categorical
136 ordinal / continuous
137 ordinal / continuous
138 ordinal / continuous
139 ordinal / continuous
140 ordinal / continuous
141 ordinal / continuous
142 ordinal / continuous
143 ordinal / continuous
144 ordinal / continuous
```

Model fitting syntax & helper functions

Fitting MVS configurations

```
# MVS application to NHANES data ----

# Libraries
library(glmnet)
library(randomForest)
library(mvs)

# Helper function to extract results
extract_results <- function(model, y_train = y_train, newx = X_test, newy = y_test) {
  pred <- predict(model, newx = newx)
  test_mse <- mean((pred - newy)^2)
  test_r2 <- 1 - test_mse / var(newy)
  coefs <- coef(model)$`Level 2`[[1]]
  cv_variances <- apply(model$`Level 1`$CVs, 2, var)
  cv_sds <- sqrt(cv_variances)
  std_coefs <- coefs[-1,] * (cv_sds / sd(y_train))

  list(
    model = model,
    coefficients = coefs,
```

```

    std_coefficients = std_coefs,
    test_MSE = test_mse,
    test_R2 = test_r2,
    cv_variances = cv_variances
  )
}

# Load data
new_nhanes_complc <- readRDS('nhanes_cc.rds')

# Set seed
set.seed(42)

# Define view indices based on source and number of dummy variables

view_ids <- c(rep(1, 29), # Sociodemographics
              rep(2, 84), # Dietary behavior
              rep(3, 23), # Medical condition
              rep(4, 6), # Sleep
              rep(5, 6), # Physical activity
              rep(6, 8) # Body measures
            )

# Create an index vector for CV folds
fold_inx <- mvs::kFolds(1:nrow(new_nhanes_complc), 10)

# Create list objects to collect extracted results
res_mvs_default <- list()
res_mvs_rf <- list()
res_mvs_repeated <- list()
res_mvs_resid <- list()

for(i in 1:10) {

  # Split into training and test data
  X_train <- model.matrix( ~ . - 1, new_nhanes_complc[fold_inx != i , -c(1, 146)])
  y_train <- new_nhanes_complc[fold_inx != i, 146]
  X_test <- model.matrix( ~ . - 1, new_nhanes_complc[fold_inx == i , -c(1, 146)])
  y_test <- new_nhanes_complc[fold_inx == i, 146]

```

```

# Fit MVS models ----

# Fit MVS default
mvs_default <- MVS(
  x = X_train,
  y = y_train,
  views = view_ids,
  progress = FALSE,
  family = "gaussian",
  std.meta = TRUE,
  std.base = FALSE
)

# Compute results
res_mvs_default[[i]] <- extract_results(mvs_default, y_train = y_train, newx = X_test, newy = y_test)

# Repeated views MVS

# Create a new predictor matrix Xrep with repeated views
Xrep_train <- cbind(X_train, X_train)
Xrep_test <- cbind(X_test, X_test)
view_ids_rep <- c(view_ids, (view_ids + 6))

# MVS residual models

# Specify type of base learner per view
type <- list(
  level1 = c(
    "StaPLR",
    "StaPLR",
    "StaPLR",
    "StaPLR",
    "StaPLR",
    "StaPLR",
    "RF",
    "RF",
    "RF",
    "RF",
    "RF",
    "RF"
  )
)

```

```

    ),
    meta = "StaPLR"
  )

  # Fit model with offset
  mvs_res_mod <- MVS(
    x = Xrep_train,
    y = y_train,
    views = view_ids_rep,
    type = type,
    progress = FALSE,
    family = "gaussian",
    offsets = list(c(1, 7), c(2, 8), c(3, 9), c(4, 10), c(5, 11), c(6, 12)),
    std.meta = TRUE
  )

  # Compute results
  res_mvs_resid[[i]] <- extract_results(mvs_res_mod, y_train = y_train, newx = Xrep_test, newy = y_test)
}

```

Helper functions

```

# Helper function to extract results
extract_results <- function(model, newx = X_test, newy = y_test) {
  pred <- predict(model, newx = newx)
  test_mse <- mean((pred - newy)^2)
  test_r2 <- 1 - test_mse / var(newy)
  coefs <- coef(model)$`Level 2`[[1]]
  cv_variances <- apply(model$`Level 1`$CVs, 2, var)

  list(
    model = model,
    coefficients = coefs,
    test_MSE = test_mse,
    test_R2 = test_r2,
    cv_variances = cv_variances
  )
}

```

```

# Helper function to summarize results
summarize_results <- function(res_list) {
  mse <- sapply(res_list, `[`, "test_MSE")
  r2 <- sapply(res_list, `[`, "test_R2")

  coefs <- do.call(cbind, lapply(res_list, `[`, "coefficients"))

  list(
    MSE_mean = mean(mse, na.rm = TRUE),
    MSE_median = median(mse, na.rm = TRUE),

    R2_mean = mean(r2, na.rm = TRUE),
    R2_median = median(r2, na.rm = TRUE),

    coef_mean = rowMeans(coefs, na.rm = TRUE),
    coef_median = apply(coefs, 1, median, na.rm = TRUE)
  )
}

# Helper function to plot coefficients
library(Matrix)
library(reshape2)
library(ggplot2)

plot_coefficients <- function(res_list, title) {
  coefs <- do.call(cbind, lapply(res_list, `[`, "coefficients"))

  coef_mat <- as.matrix(coefs)[-1, ]
  colnames(coef_mat) <- c(
    "fold 1",
    "fold 2",
    "fold 3",
    "fold 4",
    "fold 5",
    "fold 6" ,
    "fold 7",
    "fold 8",
    "fold 9",
    "fold 10"
  )
}

```

```
coef_long <- reshape2::melt(coef_mat,
                             varnames = c("view", "fold"),
                             value.name = "coefficient")

ggplot(coef_long, aes(x = view, y = coefficient)) +
  geom_boxplot() +
  geom_jitter(width = 0.15, alpha = 0.5) +
  #coord_flip() +
  labs(x = "View", y = "Coefficient", title = title) +
  theme_classic()
}
```